Technical Note

Power Electronics and Control Systems

L(RC) Filter Design Guidelines

Grid-side filters for active front ends and output filters for induction and permanent-magnet motor drives

Topologies, sizing relations, resonance placement, damping and interaction with the current control

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Abstract: Design guidelines for the passive filters of two-level voltage-source converters: the L, LC, LCL and L(RC)L filter between an active front end (AFE) and a grid of given impedance, and the LC or L(RC) output filter between an inverter and an induction motor (IM) or a permanent-magnet synchronous motor (PMSM). The note first settles the switching ripple of the converter-side inductor, with exact closed forms for sinusoidal, space-vector and discontinuous PWM, the mean and the rms ripple over the fundamental period, and a comparison with the usual $V_{dc}/(8 f_{sw} L)$ rule. It then gives, for the grid side, the sizing of the converter inductor, of the capacitor (reactive power), of the grid inductor (attenuation at the switching frequency against IEEE 519), the placement of the resonance with respect to the grid frequency, the switching frequency and the critical frequency $f_s/6$ of the digital current loop, the passive damping options (series resistor, R–C branch) with their losses, and the interaction with the grid impedance. For the motor side it covers the reflected-wave overvoltage with long cables, the dv/dt and sine filter options, the resonance window, the loaded resonance set by the high-frequency inductance of the machine (transient inductance of the IM, L_d/L_q of the PMSM), the damping, the capacitor current and voltage budget, and the consequences for the current control and for sensorless (observer-based) control. Two sizing paths with worked examples (400 V/250 kVA AFE; 110 kW IM and PMSM drives) and a bibliography close the note; the numbers are reproduced by the companion Python script.

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Nomenclature

All quantities are per phase of the star equivalent of the three-phase system; RMS values unless a hat (peak) or a "pp" (peak-to-peak) is written. Per-unit values refer to the base quantities of the converter (Z_b, L_b, C_b). The same filter symbols are used on the grid side (Fig. 1a, inductances L_1, L_2) and on the motor side (Fig. 1b, inductance L_f); the damping-network symbols are common to both.

Symbol	Description	Unit
Converter, PWM and base quantities		
V_{dc}	DC-link voltage of the two-level voltage-source converter (VSC)	V
$f_{sw},\ \omega_{sw},\ T_{sw}$	switching (carrier) frequency, $\omega_{sw} = 2\pi f_{sw}$, switching period $1/f_{sw}$	Hz, rad/s, s
$f_s,\ T_s$	update (sampling) frequency of the digital current loop, $f_s = f_{sw}$ with single update and $2f_{sw}$ with double update; $T_s = 1/f_s$; the computation and PWM delay is $1.5T_s$	Hz, s
$f_1,\ \omega_1$	fundamental frequency (grid frequency on the grid side, motor supply frequency on the motor side), $\omega_1 = 2\pi f_1$	Hz, rad/s
$f_{1,max},\ \omega_{1,max}$	highest fundamental frequency of the drive (top speed, field weakening included)	Hz, rad/s
$V,\ V_{ph}$	RMS line-to-line voltage and phase (star) voltage, $V_{ph} = V/\sqrt{3}$	V
$S_n,\ P_n$	rated apparent power of the converter, rated active power of the motor	VA, W
$I_n,\ \hat{I}_n$	rated RMS phase current and its peak, $\hat{I}_n = \sqrt{2}I_n$	A
$Z_b,\ L_b,\ C_b$	base impedance (V^2/S_n for the grid side, V_{ph}/I_n for the motor side), base inductance Z_b/ω_1 and base capacitance $1/(\omega_1 Z_b)$ at the rated fundamental frequency	Ω, H, F
m	modulation index: peak of the phase reference voltage over $V_{dc}/2$; linear range $m \le 1$ for SPWM, $m \le 2/\sqrt{3}$ for SVM/THI	–
θ	angle of the reference voltage vector; $\theta = 0$ when the phase-a voltage is at its peak, 90° at its zero crossing	rad
$v_{ref},\ v_{1,n}$	phase reference voltage and switched phase voltage with respect to the load neutral (0, $\pm V_{dc}/3$, $\pm 2V_{dc}/3$)	V
$T_1,\ T_2,\ T_0$	dwell times of the two active vectors and total dwell time of the zero vectors in one switching period; T_{0a}, T_{0b} are the parts of T_0 spent on 000 and on 111	s
SPWM, SVM, THI, DPWM	sinusoidal PWM (no zero sequence), space-vector PWM (equal split of the zero vectors), third-harmonic injection (same ripple as SVM), discontinuous PWM with 60° clamping (DPWM1)	–
k	ripple coefficient, $\Delta i = k\,V_{dc}/(f_{sw}L)$; tabulated in Table 2 as $1/x$	–
$\Delta i_{\mathrm{pp}},\ \Delta i_{\mathrm{pp},max}$	peak-to-peak switching ripple of the inductor current in one switching period, and its maximum over the fundamental period	A
$I_{rip,rms}$, HDF	RMS value of the switching ripple over the fundamental period and harmonic distortion factor of Holmes–Lipo, Eq. (2.5)	A, –
k_r	ripple allowance of the design, $\Delta i_{\mathrm{pp},max}/\hat{I}_n$	–
ϵ_L	relative fundamental voltage drop on the filter inductance(s), $\omega_1 L I_n/V_{ph}$	–
Grid-side filter (Fig. 1a)		
$L_1,\ L_2$	converter-side and grid-side filter inductances	H
$L_g,\ L_2'$	grid inductance at the point of common coupling (transformer and upstream network) and total grid-side inductance $L_2' = L_2 + L_g$	H
C_f	filter capacitance per phase, star equivalent (a delta bank uses $C_f/3$ per branch)	F
$r,\ x$	inductance ratio L_2/L_1 and capacitor fraction C_f/C_b (the capacitor reactive power is $Q_c = xS_n$)	–
$v_1,\ v_c,\ v_g$	converter, capacitor-node and grid phase voltages	V
$i_1,\ i_2$	converter-side and grid-side currents	A
$Z_1,\ Z_2,\ Z_c$	impedances of L_1, of L_2' and of the shunt network (capacitor with its damping)	Ω
$f_{res},\ \omega_{res}$	resonance frequency of the LCL filter, Eq. (3.2); f_{res}^{min}, f_{res}^{max} its values at the weakest and at the stiffest grid	Hz, rad/s
$f_2,\ \omega_2$	resonance of L_2' with C_f, $1/(2\pi\sqrt{L_2' C_f})$, the corner of the current ratio i_2/i_1	Hz, rad/s
k_a	attenuation of the filter at the switching frequency, $\lvert i_2/i_1\rvert_{\omega_{sw}}$	–
$I_{1,sw},\ I_{2,sw}$	RMS converter-side and grid-side currents of the switching-frequency cluster	A
$I_c,\ \hat{v}_{c,sw}$	RMS capacitor current and peak switching ripple of the capacitor voltage	A, V
S_{sc}, SCR	short-circuit power at the point of common coupling (PCC) and short-circuit ratio S_{sc}/S_n; I_{sc}/I_L is the ratio of IEEE 519 with I_L the maximum demand load current	VA, –
$u_k,\ S_T$	short-circuit (leakage) voltage and rating of the supply transformer	pu, VA
h, TDD	harmonic order and total demand distortion of IEEE 519	–
f_c	crossover frequency of the current loop	Hz
$f_s/6$	critical resonance frequency of the digital current loop (Parker–McGrath–Holmes)	Hz
$Y_c,\ Z_g$	converter output admittance and grid impedance of the impedance-based stability criterion	S, Ω

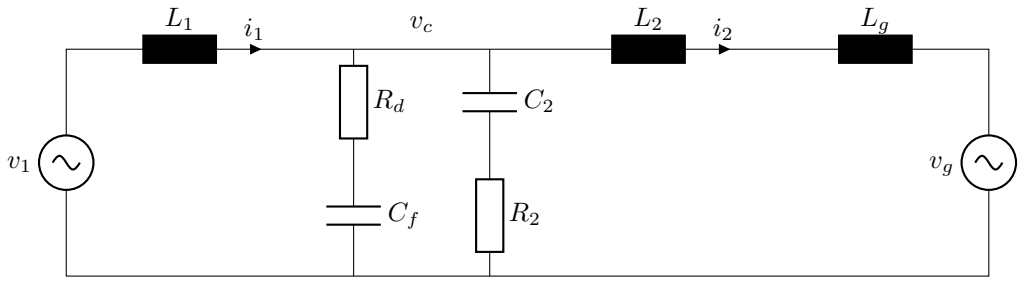
Symbol	Description	Unit
Damping networks (both sides)		
R_d	resistor in series with C_f (series damping)	Ω
$C_1,\ C_2,\ n$	main capacitor and damping-branch capacitor of the R–C network, $n = C_2/C_1$; $C_1 + C_2$ is the total capacitance	F, –
R_2	resistor of the R–C damping branch	Ω
L_p	inductance seen by the capacitor node: $L_1 \parallel L_2'$ on the grid side, $L_f \parallel L_{hf}$ on the motor side	H
$R_0,\ \omega_0$	characteristic impedance $\sqrt{L_p/C_1}$ and undamped resonance $1/\sqrt{L_pC_1}$ of the damped network (on the motor side $\omega_0 = 2\pi f_0$ with L_f)	Ω, rad/s
$\rho,\ \rho_Z,\ \rho_H,\ \rho_P$	normalised damping resistance R_2/R_0 and its closed-form optima (minimum peak impedance, minimum peak of the voltage transfer function, real pole at ω_0)	–
$\zeta,\ \zeta_m,\ Q$	damping ratio of the filter resonance, contribution of the machine to it, quality factor $1/(2\zeta)$	–
X_c	capacitor reactance at the switching frequency, $1/(\omega_{sw}C_f)$	Ω
$v_{c,sw}$	RMS switching-ripple voltage on the capacitor node	V
$P_{R_d},\ P_{R_2}$	losses of the damping resistors (fundamental plus switching terms)	W
Motor-side filter (Fig. 1b)		
$v_i,\ i_i$	inverter output voltage and current (filter input)	V, A
$v_c,\ v_s,\ i_s$	capacitor voltage, stator (motor terminal) voltage, $v_s = v_c$ without cable, and stator current	V, A
$L_f,\ R_{Lf}$	filter inductance and its series resistance	H, Ω
$L_{hf},\ R_{hf}$	inductance and resistance presented by the machine at the filter resonance: $L_{hf} = \sigma L_s = L_{ls} + L_{lr}'$ (transient inductance of the induction motor, stator and referred rotor leakage), $L_{hf} = L_d$ or L_q (PMSM); R_{hf} includes the deep-bar rotor resistance of the IM	H, Ω
e	back-EMF of the machine	V
$f_0,\ f_p$	unloaded resonance $1/(2\pi\sqrt{L_fC_f})$ and resonance with the machine connected, $f_0\sqrt{1 + L_f/L_{hf}}$	Hz
$Z_0,\ v,\ \tau,\ l$	surge impedance $\sqrt{L'/C'}$, propagation velocity, transit time l/v and length of the motor cable	Ω, m/μs, s, m
t_r	rise time (10–90 %) of the voltage edge at the cable input	s
$\Gamma_m,\ \Gamma_s$	voltage reflection coefficients at the motor and at the inverter end of the cable	–
l_{crit}	critical cable length $vt_r/2$ at which the full reflection is reached	m
$\hat{V}_m,\ OF$	peak voltage at the motor terminals and overshoot factor $\hat{V}_m/V_{dc} - 1$	V, –
$Z_m,\ C_{cable}$	high-frequency impedance of the motor seen by the cable, cable capacitance	Ω, F
G_m	equivalent conductance of the machine branch at f_p, $R_{hf}/(R_{hf}^2 + (2\pi f_p L_{hf})^2)$	S
I_c	fundamental capacitor current $\omega_1 C_f V_{ph}$	A
$\alpha_i,\ \alpha_v,\ \alpha_s$	bandwidths of the inverter-current, capacitor-voltage and stator-current loops of the cascaded control	rad/s
f_{inj}	frequency of the high-frequency injection of saliency-based sensorless control	Hz
MTPA, MTPV	maximum-torque-per-ampere and maximum-torque-per-volt trajectories of the PMSM	–

1 Scope

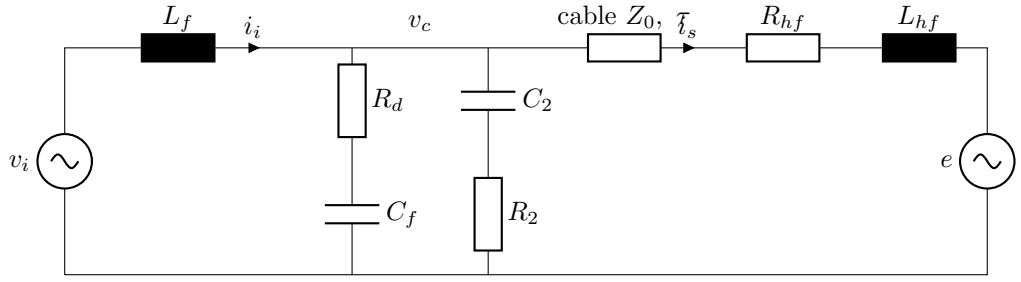
Two-level three-phase voltage-source converter (VSC) [16, 15] with DC link V_{dc}, switching frequency f_{sw}, double-edge regular-sampled PWM with single or double update. Two applications are treated with the same formulas: the grid side of an active front end (AFE), Fig. 1(a), and the output side of a motor inverter, Fig. 1(b). Formulas are quoted without derivation except where the derivation is short and not found in the references; the numbers of the worked examples are reproduced by the companion script `lrc_filter_example.py`.

2 Switching ripple of the converter-side inductor

The converter-side inductor (L_1 or L_f) is sized first from its switching ripple. With the capacitor node a stiff voltage at f_{sw} (LCL, LC) or a stiff grid (plain L), the phase current ripple over one switching period $T_{sw} = 1/f_{sw}$ is piecewise linear with slopes $(v_{1,n} - v_{ref})/L$, where $v_{1,n}$ is the converter phase voltage with respect to the load neutral (0, $\pm V_{dc}/3$, $\pm 2V_{dc}/3$) and v_{ref} the reference. Let $m = \hat{V}_{ref}/(V_{dc}/2)$ be the modulation index ($m \leq 1$



(a) Grid side: L ($C_f = 0$), LCL, L(RC)L; L_g is the grid (transformer and network) inductance.



(b) Motor side: LC, L(RC); the machine is R_{hf}, L_{hf} (σL_s for the IM, L_d/L_q for the PMSM) and the back-EMF e.

Figure 1: Filter structures. The damping is either the series resistor R_d (dashed) or the parallel branch C_2–R_2 (the "RC" of L(RC)); only one of the two is used.

for sinusoidal PWM, $m \le 2/\sqrt{3} = 1.155$ for space-vector or third-harmonic PWM). All ripple figures scale as

$$\Delta i = k\,\frac{V_{dc}}{f_{sw} L}, \tag{2.1}$$

and the coefficient k depends on the modulator, on m and on the position θ along the fundamental period. Two closed forms cover the two extremes (phase voltage at its peak, $\theta = 0$, and at its zero crossing, $\theta = 90°$):

$$k_{\mathrm{pp}}(\theta = 0) = \frac{m}{4}\Big(1 - \frac{m}{2}\Big)\ \text{(SPWM)}, \qquad \frac{m}{4}\Big(1 - \frac{3m}{4}\Big)\ \text{(SVM, THI)}, \qquad \frac{m}{2}\Big(1 - \frac{3m}{4}\Big)\ \text{(DPWM, clamped phase)}, \tag{2.2}$$

$$k_{\mathrm{pp}}(\theta = 90°) = \frac{\sqrt{3}\,m}{12} = \frac{m}{6.93} \qquad \text{(all modulators)}. \tag{2.3}$$

Eq. (2.2) follows from $\Delta i_{\mathrm{pp}} = v_{ref}\,(T_0 + |T_{0a} - T_{0b}|)/(2L)$, with $T_0 = T_{0a} + T_{0b}$ the total zero-vector time and T_{0a}, T_{0b} its split between 000 and 111 (equal for SVM, $(1-m)T_{sw}/2$ and $(1-m/2)T_{sw}/2$ for SPWM at $\theta = 0$, all on one vector for the clamped phase of DPWM); at $\theta = 0$, $T_0/T_{sw} = 1 - 3m/4$. Eq. (2.3) follows from the two active vectors of equal dwell $T_1 = T_2 = (\sqrt{3}/4)\,m\,T_{sw}$ giving $\pm V_{dc}/3$ to the phase at its zero crossing. The peak-to-peak ripple is largest at the phase peak for small m and at the zero crossing for large m (crossover at $m = 0.56$ for SVM, 0.85 for SPWM). The worst case over the whole linear range is therefore

$$\Delta i_{\mathrm{pp},max} = \frac{V_{dc}}{4\sqrt{3}\,f_{sw} L} = \frac{V_{dc}}{6.93\,f_{sw} L} \qquad \text{(SPWM, at } m = 1\text{)}, \tag{2.4}$$

$$\Delta i_{\mathrm{pp},max} = \frac{V_{dc}}{6\,f_{sw} L} \qquad \text{(SVM/THI, at } m = 2/\sqrt{3}\text{; DPWM, at } m = 2/3 \text{ and } 2/\sqrt{3}\text{)}.$$

The RMS value of the ripple over the fundamental period, needed for the inductor and capacitor losses and for the harmonic current injected into the capacitor, is given by the harmonic distortion factor of Holmes and Lipo

[15]:

$$I_{rip,rms} = \frac{V_{dc}}{24\, f_{sw} L}\sqrt{\mathrm{HDF}(m)}, \qquad \mathrm{HDF} = \begin{cases} \frac{3}{2}m^2 - \frac{4\sqrt{3}}{\pi}m^3 + \frac{9}{8}m^4 & \text{(SPWM)}, \\ \frac{3}{2}m^2 - \frac{4\sqrt{3}}{\pi}m^3 + \left(\frac{27}{16} - \frac{81\sqrt{3}}{64\pi}\right)m^4 & \text{(SVM)}. \end{cases} \tag{2.5}$$

Table 2 lists k for the maximum and the mean (over the fundamental period) peak-to-peak ripple and for the RMS ripple, computed exactly by the companion script; the closed forms (2.2)–(2.5) are reproduced to four digits.

Table 2: Ripple coefficient k of (2.1), written as $1/x$: maximum peak-to-peak over the fundamental period, mean peak-to-peak, RMS of the ripple. Two-level VSC, double-edge regular sampling, isolated neutral.

	SPWM			SVM / THI			DPWM1 (60° clamping)		
m	max pp	mean pp	RMS	max pp	mean pp	RMS	max pp	mean pp	RMS
0.4	1/12.5	1/17.6	1/67	1/14.3	1/18.7	1/68	1/7.1	1/10.4	1/35
0.6	1/9.5	1/13.7	1/52	1/11.6	1/15.3	1/55	1/6.1	1/8.8	1/30
0.8	1/8.3	1/12.1	1/44	1/8.7	1/14.4	1/49	1/6.3	1/8.9	1/31
0.9	1/7.7	1/11.4	1/41	1/7.7	1/14.0	1/47	1/6.8	1/9.4	1/32
1.0	1/6.9	1/10.9	1/37	1/6.9	1/13.5	1/45	1/6.9	1/10.4	1/35
1.1	–	–	–	1/6.3	1/13.0	1/42	1/6.3	1/11.5	1/37
1.155	–	–	–	1/6.0	1/12.8	1/40	1/6.0	1/11.9	1/38

Which ripple formula to use. The widely used $L = V_{dc}/(8 f_{sw}\Delta i_{\mathrm{pp}})$ is neither the maximum nor the mean: at $m \approx 1$ with SVM the true maximum is $V_{dc}/(6.9 f_{sw}L)$ (15 % above the rule), the worst case over m is $V_{dc}/(6 f_{sw}L)$ (33 % above, the value used by Liserre et al. [1] and Reznik et al. [2]), while the mean peak-to-peak over the fundamental is $V_{dc}/(13.5 f_{sw}L)$ (40 % below). The rule is a fair "typical period" estimate; it is not an upper bound. Recommended use: (i) size L and check the saturation current $\hat{I}_n + \Delta i_{\mathrm{pp},max}/2$ with the maximum of Table 2 at the largest m of the application ($V_{dc}/(6 f_{sw}L)$ when the converter reaches the hexagon, $V_{dc}/(7 f_{sw}L)$ for an AFE at $m \approx 1$); (ii) use the RMS column, or (2.5), for the capacitor current, the damping losses and the harmonic current at f_{sw}; (iii) with discontinuous PWM at the same f_{sw} the ripple is up to twice that of SVM at low m, so compare at equal switching losses (f_{sw} raised by 1.5).

3 Filter structures and transfer functions

With $Z_1 = sL_1$, $Z_2 = s(L_2 + L_g)$ and Z_c the impedance of the shunt network of Fig. 1(a), and the grid voltage a short at the switching harmonics,

$$\frac{i_1}{v_1} = \frac{Z_c + Z_2}{Z_1 Z_2 + Z_c(Z_1 + Z_2)}, \qquad \frac{i_2}{v_1} = \frac{Z_c}{Z_1 Z_2 + Z_c(Z_1 + Z_2)}, \qquad \frac{i_2}{i_1} = \frac{Z_c}{Z_c + Z_2}. \tag{3.1}$$

For the undamped LCL ($Z_c = 1/(sC_f)$), writing $L_2' = L_2 + L_g$,

$$\frac{i_2}{v_1} = \frac{1}{s^3 L_1 L_2' C_f + s(L_1 + L_2')}, \qquad \omega_{res}^2 = \frac{L_1 + L_2'}{L_1 L_2' C_f}, \qquad \frac{i_2}{i_1} = \frac{1}{1 + s^2 L_2' C_f}, \qquad \omega_2^2 = \frac{1}{L_2' C_f}. \tag{3.2}$$

At the switching frequency the attenuation of the LCL with respect to its own converter current is

$$k_a = \left|\frac{i_2}{i_1}\right|_{\omega_{sw}} = \frac{1}{\omega_{sw}^2 L_2' C_f - 1} \approx \left(\frac{f_2}{f_{sw}}\right)^2, \qquad \text{equivalently [1]} \quad \frac{i_2(\omega_{sw})}{i_L(\omega_{sw})} = \frac{1}{\left|1 + r\,(1 - \omega_{sw}^2 L_1 C_f)\right|}, \quad r = \frac{L_2}{L_1}, \tag{3.3}$$

the second form being referred to the current $i_L = v_1/(j\omega_{sw}L_1)$ of the plain L filter; the two coincide where C_f is a short at f_{sw}. The damped networks are $Z_c = R_d + 1/(sC_f)$ (series resistor) and $Z_c = \frac{1}{sC_1} \parallel \left(R_2 + \frac{1}{sC_2}\right)$



(R–C branch, with $C_1 + C_2$ the total capacitance). The series resistor turns the -60 dB/decade roll-off of i_2/v_1 into -40 dB/decade above $1/(2\pi R_d C_f)$ and degrades (3.3) to $k_a = \sqrt{R_d^2 + X_c^2}/\sqrt{R_d^2 + (\omega_{sw} L_2' - X_c)^2}$, $X_c = 1/(\omega_{sw} C_f)$; the R–C branch keeps the roll-off and is analysed in [14, 11, 12, 13]. The motor-side LC filter of Fig. 1(b) obeys the same algebra with $Z_1 = sL_f$ and $Z_2 = R_{hf} + sL_{hf}$ (the back-EMF is a short at the ripple frequencies, the cable adds its capacitance to C_f):

$$\frac{v_c}{v_i} = \frac{Z_c \parallel Z_2}{Z_1 + Z_c \parallel Z_2}, \qquad f_0 = \frac{1}{2\pi\sqrt{L_f C_f}}, \qquad f_p = f_0\sqrt{1 + \frac{L_f}{L_{hf}}}, \qquad \left|\frac{v_c}{v_i}\right|_{f_{sw}} \approx \frac{1}{(f_{sw}/f_p)^2 - 1}, \tag{3.4}$$

where f_0 is the unloaded resonance and f_p the resonance with the machine connected.

4 Grid-side filter of an active front end

4.1 Design inputs

Rated power S_n, voltage V, frequency f_1, DC-link voltage V_{dc} (boost operation requires $V_{dc} \geq \sqrt{2}V(1+\epsilon_L)$, i.e. $m \leq 2\sqrt{2}V_{ph}(1+\epsilon_L)/V_{dc}$ with ϵ_L the filter drop), f_{sw} and f_s, the current-feedback choice (converter current i_1 or grid current i_2), the range of the grid impedance at the point of common coupling (PCC), and the harmonic limits. The grid inductance per phase, transformer included, is

$$L_g = \frac{V^2}{\omega_1 S_{sc}} = \frac{u_k V^2}{\omega_1 S_T} + L_g^{MV}, \qquad \frac{S_{sc}}{S_n} = \frac{I_{sc}}{I_L} \text{ (short-circuit ratio)}, \tag{4.1}$$

to be bracketed between the stiffest ($S_{sc,max}$) and the weakest ($S_{sc,min}$) expected PCC. The limits of IEEE 519 [17] at the PCC depend on I_{sc}/I_L: for $I_{sc}/I_L < 20$ the individual current harmonics are limited to 4 % ($h < 11$), 2 % ($11 \leq h < 17$), 1.5 % ($17 \leq h < 23$), 0.6 % ($23 \leq h < 35$) and 0.3 % ($35 \leq h \leq 50$) of I_L, with TDD 5 %; the switching-frequency cluster of an AFE lies above $h = 50$, where the 0.3 % value is used as design practice. IEC 61800-3 [18] sets the EMC environment (C1–C4) and refers to the IEC 61000-3 series for the low-frequency emission.

4.2 Converter-side inductor L_1

From Section 2, with the ripple allowance $k_r = \Delta i_{\mathrm{pp},max}/\hat{I}_n$ (15–30 % is usual; a smaller value increases L_1, its losses and the voltage drop, a larger one the switching losses and the capacitor current),

$$L_1 \geq \frac{k\, V_{dc}}{f_{sw}\, k_r \hat{I}_n}, \qquad k = \frac{1}{7} \text{ (SVM, } m \approx 1), \quad \frac{1}{6} \text{ (worst case)}, \tag{4.2}$$

and the drop $\epsilon_L = \omega_1(L_1 + L_2)I_n/V_{ph}$ (typically 0.05–0.15 pu in total) must be compatible with V_{dc}. The inductor carries I_n plus the ripple, must not saturate at $\hat{I}_n + \Delta i_{\mathrm{pp},max}/2$ (plus the overload and the inrush at pre-charge), and its core loss is set by the ripple RMS of Table 2.

4.3 Capacitor C_f

The capacitor is bounded by its reactive power at the fundamental, which the converter must compensate (and which raises the no-load current),

$$C_f \leq x\, C_b = \frac{x\, S_n}{\omega_1 V^2}, \qquad Q_c = \omega_1 C_f V^2 = x S_n, \qquad x \leq 0.05 \text{ [1]}, \tag{4.3}$$



(C_f is the star value; a delta bank uses $C_f/3$ per branch). A larger C_f lowers L_2 for a given attenuation and lowers f_{res}; with $x = 2$–$5\,\%$ the capacitor is the cheapest element and is normally taken at its upper bound. The capacitor must withstand $1.1V$ plus the switching ripple voltage $\hat{v}_{c,sw} \approx I_{rip,rms}\sqrt{2}/(\omega_{sw}C_f)$, carry $I_c = \sqrt{(\omega_1 C_f V_{ph})^2 + I_{rip,rms}^2}$ and be rated for the harmonic background of the grid (5th, 7th) that it absorbs through L_2'.

4.4 Grid-side inductor L_2 and attenuation

The converter-side current at the switching frequency is estimated from the ripple RMS of Table 2 (assigning the whole ripple to the first carrier group is conservative by about $15\,\%$; the sideband spectrum is in [15]); with the admissible grid harmonic $I_{2,sw} \le 0.003 I_n$ the required attenuation is $k_a = I_{2,sw}/I_{1,sw}$ and, from (3.3) with $L_g = 0$ (worst case: L_g only adds attenuation),

$$L_2 \ge \frac{1 + 1/k_a}{\omega_{sw}^2 C_f}, \qquad r = \frac{L_2}{L_1} = 0.2\text{–}1. \tag{4.4}$$

Values of r above unity give little further attenuation for the volume [1, 3]; the grid inductor may be the leakage of the dedicated transformer or a line reactor. If the attenuation cannot be met, raise C_f up to its bound, then L_1 (which lowers $I_{1,sw}$ linearly), then f_{sw}.

4.5 Resonance placement

With L_g between L_g^{min} and L_g^{max} the resonance (3.2) sweeps the band $[f_{res}^{min}, f_{res}^{max}]$, f_{res}^{min} at the weakest grid. The constraints are

$$10 f_1 \le f_{res}^{min}, \qquad f_{res}^{max} \le \frac{f_{sw}}{2}\ [1], \qquad f_{res} \notin \Big[0.8, 1.2\Big]\frac{f_s}{6}\ [6]. \tag{4.5}$$

The lower bound keeps the resonance away from the low-order grid harmonics and from the current-loop crossover f_c (design $f_c \le f_{res}/3$, with $f_c \approx f_{sw}/20 \ldots f_{sw}/10$ for a PI in the dq frame [7]); the upper bound preserves the attenuation at f_{sw}. The third constraint comes from the discrete-time analysis of the current loop with the computation and PWM delay ($1.5T_s$): with converter-current feedback the LCL resonance is stable without any damping only for $f_{res} > f_s/6$, with grid-current feedback only for $f_{res} < f_s/6$, and at $f_{res} = f_s/6$ no controller gain stabilises the loop [6]. Place the whole band $[f_{res}^{min}, f_{res}^{max}]$ on the side consistent with the feedback variable, or provide damping (passive, Section 4.6, or active) whenever the band crosses or approaches $f_s/6$ or when the grid impedance is uncertain [8, 5].

4.6 Passive damping

Series resistor. The classical choice [1] is one third of the capacitor reactance at the resonance,

$$R_d = \frac{1}{3\omega_{res}C_f}, \qquad \zeta \approx \frac{R_d}{2}\sqrt{\frac{C_f}{L_p}} \approx \frac{1}{6}, \qquad L_p = \frac{L_1 L_2'}{L_1 + L_2'}, \qquad P_{R_d} = 3R_d\big[(\omega_1 C_f V_{ph})^2 + I_{rip,rms}^2\big], \tag{4.6}$$

where the loss has a fundamental term (the whole capacitor current flows in R_d) and a switching term; both are usually 0.1–0.3 % of S_n [4]. The price is the attenuation penalty of Section 3.

R–C branch. Splitting the capacitance into C_1 and $C_2 = nC_1$ with R_2 in series with C_2 keeps the -60 dB/decade



slope. With $R_0 = \sqrt{L_p/C_1}$ and $\rho = R_2/R_0$, the three closed-form choices of [14] are

$$\rho_Z = \sqrt{\frac{(2+n)(4+3n)}{2n^2(4+n)}} \qquad \text{minimum peak of the output impedance [11],} \tag{4.7}$$

$$\rho_H = \frac{\sqrt{(1+n)(2+n)/2}}{n} \qquad \text{minimum peak of } |v_c/v_1|,$$

$$\rho_P = \frac{2+n}{2n} \qquad \text{real pole at } \omega_0 = 1/\sqrt{L_pC_1},$$

with $\zeta = n/(2\sqrt{2(2+n)})$ for ρ_P (0.20 for $n = 1$, 0.35 for $n = 2$) and a peak of $|v_c/v_1|$ not lower than $(2+n)/n$. On the AC grid the branch carries the fundamental capacitive current of C_2, so

$$P_{R_2} = 3R_2\frac{(\omega_1 C_2 V_{ph})^2}{1+(\omega_1 C_2 R_2)^2} + 3\frac{v_{c,sw}^2}{R_2}, \qquad v_{c,sw} \approx \frac{I_{rip,rms}}{\omega_{sw}C_1}, \tag{4.8}$$

which limits n to 1–2 and makes the loss comparable with the series resistor [4]. Both networks are to be checked at f_{res}^{min} and f_{res}^{max}; in modern AFEs passive damping is kept small as a robustness backstop and the bulk of the damping is active (capacitor-current or capacitor-voltage feedback [7, 5, 10]).

4.7 Interaction with the grid impedance and stability

The grid sees the converter as a current source with an output admittance $Y_c(s)$ shaped by the current controller; stability of the interconnection is that of the loop $Z_g(s)Y_c(s)$ [9]. With the digital delay the real part of Y_c becomes negative between about $f_s/6$ and $f_s/2$ for converter-current control, so any resonance of Z_g with the filter falling in that band is destabilised, whatever the gains; passive damping and the choice of f_{res} of Section 4.5 move the resonance out of the band or make Y_c passive there. Practical rules: (i) evaluate f_{res} and the phase margin at both ends of the L_g range; (ii) a weak grid (SCR < 3) also demands a slow PLL ($\leq 20\,\text{Hz}$) and a reduced current-loop bandwidth; (iii) the background distortion of the grid is amplified by the LCL between f_2 and f_{res}, which is why $f_{res} \geq 10f_1$ and why the capacitor must be rated for the 5th/7th harmonics; (iv) verify the design with the transformer leakage and the cable of the actual installation, not with $L_g = 0$ only.

4.8 Sizing path and worked example

Step	Relation	Example: 400 V, 250 kVA, 50 Hz, $V_{dc} = 700\,\text{V}$, $f_{sw} = 4\,\text{kHz}$, $f_s = 8\,\text{kHz}$, SVM, i_1 feedback
G1	base values, L_g range from (4.1)	$I_n = 361\,\text{A}$, $Z_b = 0.64\,\Omega$, $L_b = 2.04\,\text{mH}$, $C_b = 4.97\,\text{mF}$; $S_{sc} =$ 5–20 MVA ($I_{sc}/I_L = 20\ldots80$) $\Rightarrow L_g =$ 25–102 µH
G2	L_1 from (4.2), $k_r = 0.25$ at the rated m	$m = 0.98$: $k = 1/7.07$, $L_1 \geq 194\,\mu\text{H} \Rightarrow L_1 = 200\,\mu\text{H}$ (9.8 %): $\Delta i_{pp,max} = 124\,\text{A}$ (24 %), mean 64 A, $I_{rip,rms} = 19.2\,\text{A}$; worst case $V_{dc}/(6f_{sw}L_1) = 146\,\text{A}$, rule $V_{dc}/(8f_{sw}L_1) = 109\,\text{A}$
G3	C_f from (4.3), $x = 0.05$	$C_f \leq 249\,\mu\text{F} \Rightarrow C_f = 250\,\mu\text{F}$ star (83 µF delta), $Q_c = 12.6\,\text{kvar}$
G4	k_a and L_2 from (4.4)	$I_{2,sw} \leq 0.003I_n = 1.08\,\text{A}$, $I_{1,sw} \approx 19.2\,\text{A} \Rightarrow k_a \leq 0.056$ ($-25\,\text{dB}$), $L_2 \geq 119\,\mu\text{H} \Rightarrow L_2 = 125\,\mu\text{H}$ ($r = 0.62$), $k_a = 0.053$, $I_{2,sw} = 1.03\,\text{A}$ (0.28 %); $L_1 + L_2 = 16\,\%$
G5	f_{res} from (3.2), band (4.5)	$f_{res} = 1148\,\text{Hz}$ ($L_g = 0$), 976–1086 Hz over the L_g range; $10f_1 =$ 500 Hz, $f_{sw}/2 = 2\,\text{kHz}$; $f_s/6 = 1333\,\text{Hz}$: the band is below $f_s/6$ with i_1 feedback $\Rightarrow$ damping required (or i_2 feedback)
G6	damping (4.6) or (4.7)–(4.8)	series: $R_d = 185\,\text{m}\Omega$, $\zeta \approx 0.17$, $P = 387\,\text{W}$ (0.15 %), k_a degraded to 0.082; R–C with $n = 1$: $C_1 = C_2 = 125\,\mu\text{F}$, $L_p = 77\,\mu\text{H}$, $R_0 =$ $0.78\,\Omega$, $\rho_P = 1.5 \Rightarrow R_2 = 1.18\,\Omega$, $\zeta = 0.20$, $P = 290+95 = 385\,\text{W}$ (0.15 %), slope preserved
G7	ratings	L_1: 361 A, saturation $\geq 583\,\text{A}$, ripple 19 A RMS at 4 kHz; L_2: line reactor 361 A; C_f: $\geq 440\,\text{V}$ AC, $I_c \approx 26\,\text{A}$
G8	control checks	$f_c \leq f_{res}^{min}/3 \approx 320\,\text{Hz}$ ($\approx f_{sw}/12$); verify phase margin at L_g^{min} and L_g^{max} with the delay $1.5T_s$; PLL $\leq 20\,\text{Hz}$ at $S_{sc,min}$



The example shows the typical outcome at 4 kHz: a 0.3 % target at the switching cluster needs $L_1+L_2 \approx 0.15$ pu with the capacitor at its 5 % bound, and the resonance lands at $f_{res}/f_{sw} \approx 0.25$–$0.3$, i.e. below the critical $f_s/6$ with double update. Raising f_{sw} to 8–10 kHz halves the inductances and moves f_{res} above $f_s/6$ at the same x; with grid-current feedback the present placement is instead the inherently damped one.

5 Output filters between inverter and motor

5.1 Why: dv/dt, reflected waves and insulation stress

A PWM edge with rise time t_r travels along the motor cable (surge impedance $Z_0 = \sqrt{L'/C'} \approx 20$–$80\,\Omega$, velocity $v \approx 150$–200 m/µs, transit time $\tau = l/v$) and is reflected at the motor with coefficient $\Gamma_m = (Z_m - Z_0)/(Z_m + Z_0)$, close to +1 because the high-frequency impedance of the winding is $\gg Z_0$ (0.8–0.95 for motors below 100 kW), and at the inverter with $\Gamma_s \approx -1$. The terminal voltage reaches $V_{dc}(1+\Gamma_m)$ when the edge is shorter than the round trip [23, 24]:

$$l_{crit} = \frac{v\,t_r}{2}, \qquad \hat{V}_m \le V_{dc}\Big[1 + \Gamma_m \min\Big(1, \frac{l}{l_{crit}}\Big)\Big], \tag{5.1}$$

the second expression being the design envelope of [25] (the lossless-line solution for a linear edge lies below it; real edges and cable losses smooth it). Double pulsing and polarity reversal within the cable ringing period 4τ raise the peak up to $3V_{dc}$ [26]; the high-frequency models needed for prediction are in [27]. With IGBT edges of 0.1–0.3 µs, l_{crit} is 10–30 m and any industrial cable gives the full $2V_{dc}$. The limits are those of the motor insulation: NEMA MG 1 Part 31 [22] (definite-purpose inverter-fed motors) allows $3.1\,V_{rated}$ peak with $t_r \ge 0.1$ µs for ≤ 600 V motors, Part 30 (general purpose) 1000 V peak and $t_r \ge 2$ µs; IEC 60034-18-41 [20] qualifies the insulation by stress category (overshoot factor and impulse rise time), IEC TS 60034-25 [21] gives the application guide and IEC TS 61800-8 [19] the method to specify the voltage at the power interface. The common-mode dv/dt (bearing currents, EMI) is a separate issue, mitigated by common-mode chokes and not covered here.

5.2 Filter options

- **Output reactor** (1–3 %): slows the edge seen by the cable only through the cable capacitance, $t_r \approx 2.2\sqrt{LC_{cable}}$-like; mostly limits the capacitive charging current and the inverter stress; insufficient alone above 50–100 m.
- dv/dt **filter** (LC, $f_0 = 5$–$10\,f_{sw}$, damped): limits the edge at the filter output to $t_r \approx 0.3/f_0$ (undamped 10–90 % rise; 0.3–0.5 with damping). From (5.1), for an admissible overshoot $OF = \hat{V}_m/V_{dc} - 1$ and cable length l: $l_{crit} \ge l\,\Gamma_m/OF$, i.e. $t_r \ge 2l\Gamma_m/(v\,OF)$. The capacitor is charged and discharged twice per switching period, so irrespective of the damping resistor the loss is bounded by $P \approx 3 f_{sw} C_f V_{dc}^2$: C_f must stay at 0.1–0.5 µF and the filter cannot be made slow. Damping $\zeta = 0.3$–0.5 with R_d in series with C_f or a resistor across L_f (carries no fundamental); clamping diodes to the DC bus recover the energy [28, 25].
- **Sine filter** (LC or L(RC), $f_0 \ll f_{sw}$): sinusoidal motor voltage, no reflection problem at any length (the edge is $\sim 1/(2f_0)$), lower motor noise and harmonic losses, standard motors usable, but fundamental drop, capacitor current, reduced control bandwidth and field-weakening range [29, 28, 32]. The remainder of this section is on the sine filter.
- **Motor-terminal RC** ($R = Z_0$, $C \approx 3t_r/Z_0$): matches the cable at the motor end [24]; used when the filter cannot be at the inverter.



5.3 Sine filter: resonance window

The unloaded resonance f_0 must sit between the highest fundamental and its low-order harmonics on one side, and the switching frequency on the other:

$$(8\dots10)\,f_{1,max} \le f_0 \le \frac{f_{sw}}{4\dots3}, \qquad \left|\frac{v_c}{v_i}\right|_{f_{sw}} \approx \left(\frac{f_p}{f_{sw}}\right)^2 = -20\text{–}-28\,\text{dB}. \tag{5.2}$$

The lower bound keeps the 5th and 7th harmonics generated by the dead time and by the inverter nonlinearity (at $5f_1$, $7f_1$, up to 1.4 kHz for a 200 Hz machine) away from the resonance, where they would be amplified by $Q = 1/(2\zeta)$, and keeps the current-loop crossover below $f_0/3$. The window exists only if $f_{sw} \ge (24\dots40)\,f_{1,max}$: a 50 Hz induction motor in field weakening up to 75 Hz is served by $f_{sw} = 4$ kHz, a high-speed or high-pole PMSM at 200 Hz needs $f_{sw} \ge$ 6–8 kHz or a relaxed attenuation. This, with the capacitor current, is the main reason why sine filters are heavier for PMSM drives.

5.4 Sine filter: inductor, capacitor, loaded resonance

Inductor. Sized by the fundamental drop at $f_{1,max}$ and checked for the inverter ripple,

$$L_f \le \frac{\epsilon_L V_{ph}}{\omega_{1,max} I_n}, \quad \epsilon_L = 0.05\text{–}0.10; \qquad \Delta i_{\text{pp},max} = \frac{V_{dc}}{6 f_{sw} L_f} \quad \text{(the inverter reaches } m = 2/\sqrt{3} \text{ at } f_{1,max}\text{)}, \tag{5.3}$$

With $L_f = 3\text{–}6\,\%$ the inverter current ripple is 25–45 % of $\hat{I}_n$: it flows entirely into C_f, the motor sees none of it, but the inverter must be rated for the peak and for the switching losses, and the inductor core for the ripple RMS of Table 2. The drop reduces the voltage available to the machine by $\omega_1 L_f I_n$ (vectorially): it must be budgeted in V_{dc} or accepted as an earlier field-weakening onset.

Capacitor. From the chosen f_0 and the fundamental capacitive current,

$$C_f = \frac{1}{(2\pi f_0)^2 L_f}, \qquad I_c = \omega_{1,max} C_f V_{ph} \le (0.1\dots0.2)\,I_n, \qquad \boldsymbol{i}_i = \boldsymbol{i}_s + j\omega_1 C_f \boldsymbol{v}_c, \quad \boldsymbol{v}_i = \boldsymbol{v}_c + (R_{Lf} + j\omega_1 L_f)\,\boldsymbol{i}_i, \tag{5.4}$$

the last two relations giving the inverter current and voltage that set the inverter rating and shift the torque-maximising trajectories of the machine [32]. The capacitor is rated for $1.1V$ plus the switching ripple $\hat{v}_{c,sw} \approx \sqrt{2} I_{rip,rms}/(\omega_{sw} C_f)$ and for $\sqrt{I_c^2 + I_{rip,rms}^2}$.

Loaded resonance. With the machine connected the resonance moves from f_0 to $f_p = f_0\sqrt{1 + L_f/L_{hf}}$ of (3.4), where L_{hf} is the inductance the machine presents at kilohertz frequencies: the transient (total leakage) inductance $\sigma L_s = L_{ls} + L'_{lr}$ of the induction motor, 0.15–0.25 pu, or L_d, L_q of the PMSM, 0.1–0.3 pu for surface magnets and 0.3–0.8 pu for interior magnets with $L_q > L_d$ and a strong dependence on i_q (saturation). The shift is 4–20 % and must be included in (5.2); for an IPM it differs between the d and q axes, so the plant seen by the current controller is anisotropic. The damping contributed by the machine is small because $L_{hf} \gg L_f$ decouples it from the capacitor node: the machine branch acts at f_p as a conductance $G_m = R_{hf}/(R_{hf}^2 + (2\pi f_p L_{hf})^2)$ and gives $\zeta_m = \frac{1}{2} G_m \sqrt{L_p/C_f}$ with $L_p = L_f \parallel L_{hf}$, i.e. a few 10^{-3} even with the deep-bar rotor resistance of the IM (0.05–0.15 pu at 1 kHz), and 10^{-4} for the PMSM, whose rotor has no cage. The resonance therefore has $Q \approx$ 10–20, set by the inductor losses, unless it is damped explicitly.



5.5 Sine filter: damping

The same two passive networks of Section 4.6 apply with $\omega_{res} \to 2\pi f_p$ and $L_p = L_f \parallel L_{hf}$; on the motor side the fundamental reaches $f_{1,max}$ and the capacitor is a larger fraction of C_b, so the fundamental term dominates the loss of both networks and the R–C branch, which adds $C_2 = nC_f$ to the capacitance, costs typically twice the series resistor. The series resistor $R_d = 1/(3 \cdot 2\pi f_p C_f)$ ($\zeta \approx 0.17$) is the usual passive choice at 0.5–1 % of P_n; it also reduces the attenuation by the factor $\sqrt{1 + (2\pi f_{sw} R_d C_f)^2}$. Active damping (a virtual resistor through capacitor-voltage or capacitor-current feedback, or the cascaded control of the next section) is lossless and is preferred where the control has the necessary measurements; the passive resistor is then kept small as a backstop for the V/f mode and for the resonance excited by the dead-time harmonics.

5.6 Consequences for the control, IM versus PMSM

With the filter the inverter current i_i and voltage v_i are no longer the stator quantities: $i_s = i_i - C_f\, dv_c/dt$ and $v_s = v_c = v_i - (R_{Lf} + L_f\, d/dt)\, i_i$. A vector controller that ignores the filter commits a steady-state error of 5–15 % on the voltage (the drop on L_f) and of 5–20 % on the current (the capacitor current), and has the undamped LC resonance inside its loop. The structures that work are:

- **Cascaded control** [30]: inner loop on i_i (plant L_f, bandwidth $\alpha_i \le 2\pi f_{sw}/10$), middle loop on v_c (plant C_f, $\alpha_v \approx \alpha_i/3 \ldots \alpha_i/5$), outer loop on i_s (plant L_{hf}, $\alpha_s \approx \alpha_v/3 \ldots \alpha_v/5$); the inner loops damp the resonance actively. The stator-current bandwidth ends at $f_p/5$–$f_p/3$, i.e. 100–400 Hz instead of the 0.5–1 kHz of a filterless drive, and the speed loop is correspondingly slower; this is the price for servo-like PMSM applications.
- **Full-order observer** of the filter and machine states from i_i (and v_i reference), which restores i_s, v_s and the flux without extra sensors [30, 31]; the observer must include L_f, C_f and the correct L_{hf} (for the IPM, L_d and L_q with their saturation).
- **Direct measurement** of i_s or v_c after the filter when the cabinet layout allows it.

Sensorless control. Back-EMF (fundamental-model) observers of the PMSM need v_s: using v_i instead of v_c injects the L_f drop directly into the position estimate, so the filter model must be in the observer [31]; the same holds for the IM flux observer [30]. Saliency-based high-frequency injection (f_{inj} = 0.5–2 kHz) is attenuated and phase-shifted by the filter and, near f_p, the response is dominated by the filter rather than by the saliency: keep $f_{inj} \le f_p/3$ with the filter transfer function compensated, or measure i_s directly; many drives disable injection with a sine filter.

IM versus PMSM, summary. (i) L_{hf}: σL_s for the IM, L_d/L_q for the PMSM, larger and anisotropic for the IPM, so the loaded resonance shift and the plant differ per axis. (ii) The IM rotor cage adds a lossy branch but, as shown, the damping it gives to the filter is marginal; the PMSM gives none. (iii) The PMSM usually has the higher $f_{1,max}$, which narrows the window (5.2) and raises the required f_{sw}. (iv) The filter drop and the capacitor current change the voltage and current limits: for the PMSM they shift the MTPA/MTPV trajectories and anticipate the field weakening [32]; for the IM they reduce the maximum torque in field weakening. (v) The benefit differs too: for the PMSM the sinusoidal current removes the magnet eddy-current losses and the demagnetisation risk from the switching ripple at high temperature; for the IM it removes the rotor-bar harmonic losses and the acoustic noise.

5.7 Sizing path and worked example

Step	Relation	IM 110 kW, 4-pole, 400 V, 195 A, $f_{1,max}=75$ Hz, $\sigma L_s=0.18$ pu, $f_{sw}=4$ kHz	PMSM (IPM) 110 kW, 400 V, 180 A, $f_{1,max}=200$ Hz, $L_q\approx 0.3$ pu, $f_{sw}=8$ kHz
M1	cable (5.1): $v=150$ m/µs, $t_r=0.2$ µs, $l=150$ m, $\Gamma_m=0.9$	$l_{crit}=15$ m $\Rightarrow \hat{V}_m\approx 1.9V_{dc}=1235$ V with $V_{dc}=650$ V: at the MG 1-31 limit (1240 V), above MG 1-30 (1000 V) $\Rightarrow$ filter needed; a dv/dt filter for $OF\le 0.2$ would need $t_r\ge 9$ µs ($f_0\le 33$ kHz) with $C_f\le 0.22$ µF (loss bound 1.1 kW at 4 kHz); the sine filter is chosen	
M2	window (5.2)	$8f_{1,max}=600$ Hz $\le f_0\le f_{sw}/4=1$ kHz	1.6 kHz $\le f_0\le$ 2 kHz (infeasible at 4 kHz)
M3	L_f from (5.3)	$\epsilon_L\le 0.08$ at 75 Hz $\Rightarrow L_f\le 201$ µH; $L_f=230$ µH (6.1 %): drop 6.1 % at 50 Hz, 9.2 % at 75 Hz; $\Delta i_{\mathrm{pp},max}=118$ A (43 %), RMS 17.8 A	$L_f\le 82$ µH would give 65 % ripple $\Rightarrow L_f=150$ µH (3.7 %): drop 14.7 % at 200 Hz (budgeted in V_{dc}); $\Delta i_{\mathrm{pp},max}=90$ A (35 %), RMS 13.6 A
M4	C_f from (5.4)	$f_0\to 900$ Hz: $C_f=140$ µF (5.2 % C_b), $f_0=887$ Hz; $I_c=10.2/15.2$ A at 50/75 Hz (8 %)	$f_0\to 1.8$ kHz: $C_f=50$ µF (2 % C_b), $f_0=1838$ Hz; $I_c=14.5$ A at 200 Hz (8 %)
M5	loaded resonance (3.4)	$L_{hf}=0.68$ mH: $f_p=1.16f_0=1026$ Hz (at the window edge), $\lvert v_c/v_i\rvert_{f_{sw}}=0.070$ (−23 dB), $\zeta_m\approx 0.003$	$L_{hf}=1.23$ mH: $f_p=1.06f_0=1947$ Hz, $\lvert v_c/v_i\rvert_{f_{sw}}=0.063$ (−24 dB), $\zeta_m\approx 10^{-4}$
M6	damping	series $R_d=0.37\,\Omega$ ($\zeta\approx 0.17$): 607 W (0.55 %) at 75 Hz; R–C $n=1$: +140 µF, $R_2=1.66\,\Omega$, 1.19 kW (1.1 %)	series $R_d=0.55\,\Omega$: 648 W (0.6 %) at 200 Hz; R–C $n=1$: +50 µF, $R_2=2.45\,\Omega$, 1.55 kW (1.4 %) $\Rightarrow$ active damping preferred
M7	control, Section 5.6	$\alpha_i\le 2\pi\cdot 400$ Hz, stator-current loop $\le f_p/5\approx 200$ Hz, $f_{inj}\le 340$ Hz	$\alpha_i\le 2\pi\cdot 800$ Hz, stator-current loop ≤ 390 Hz, $f_{inj}\le 650$ Hz; $L_d\neq L_q$ in the observer

The two columns show the trade-off: the induction motor at 4 kHz fits the window with a 6 % inductor and a 5 % capacitor, the PMSM needs 8 kHz and still pays a 15 % drop at its top speed or a larger ripple; in both cases the passive damping costs 0.5–1.5 % of the rated power, which is why the resonance is damped actively whenever the controller has the filter in its model. The dead-time harmonics must be checked against f_p at every operating speed (the 7th of 75 Hz is 525 Hz, that of 200 Hz is 1.4 kHz, both below f_p with margin), and the saturation of L_q at rated i_q shifts f_p of the IPM upwards by a few percent more.

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